# NK Annotation Biomarker and Agent Logic Report

Current pipeline summary based on local annotation-agent code and marker reference files.

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## 1. What This Report Documents

• The pan-NK marker panel used to evaluate broad NK identity.

• The non-NK lineage marker panels used as sanity checks for T, B, myeloid, erythroid, stromal, and epithelial signals.

• The Amina unified NK taxonomy marker reference used to assign NK subtype and state labels.

• How the annotation agent combines cluster-vs-rest DE, taxonomy marker hits, metadata context, pairwise DE, and guardrails to produce final labels.

### Source files

| Component | Local source |
|---|---|
| Pan-NK and non-NK marker panels | nk_project/annotation_agent/marker_knowledge.py |
| Amina taxonomy marker reference | nk_project/annotation_agent/references/FINAL_UNIFIED_NK_TAXONOMY_REFERENCE_nolayer.md |
| Taxonomy parsing and scoring | nk_project/annotation_agent/taxonomy_reference.py |
| Evidence construction and sanity checks | nk_project/annotation_agent/evidence.py |
| Annotation and review prompts | nk_project/annotation_agent/prompts.py |
| Decision normalization/guardrails | nk_project/annotation_agent/graph.py |

## 2. Core Pan-NK Biomarkers

These genes are used as a broad NK identity panel. In this NK-enriched analysis, negative cluster-vs-rest DE for pan-NK genes means lower than other NK-rich clusters, not necessarily absent NK identity.

| Panel | Genes |
|---|---|
| Pan-NK markers | NCR1, KLRD1, KLRF1, NCAM1, EOMES, TBX21, NKG7, GNLY, PRF1, GZMB, GZMA, CST7 |

## 3. Non-NK Biomarker Panels

These marker groups are used to detect possible non-NK lineage signal. They are interpreted together with pan-NK support and the NK-enriched background.

| Lineage/program | Markers |
|---|---|
| T | CD3D, CD3E, CD3G, TRAC, TRBC1, TRBC2 |
| B | MS4A1, CD79A, CD79B, CD19, BANK1 |
| Myeloid | LYZ, LST1, S100A8, S100A9, CST3 |
| Erythroid | HBB, HBA1, HBA2, AHSP |
| Stromal_Epithelial | COL1A1, DCN, LUM, EPCAM, KRT8, KRT18, KRT19 |

## 4. Marker Programs Used For Context

Program-level hits are derived from the top positive and negative cluster DE genes. They help the agent recognize biology such as cytotoxicity, tissue/regulatory chemokine programs, proliferation, cytokine/interferon response, and contamination-like lineages.

| Program | Markers |
|---|---|
| NK cytotoxic | NKG7, GNLY, PRF1, GZMB, GZMH, GZMA, CST7, FGFBP2, KLRF1, FCGR3A |
| NK tissue/regulatory/chemokine | XCL1, XCL2, KLRC1, KLRC2, KLRB1, CXCR6, ITGAE, ZNF683, CCL3, CCL4, CCL5 |

| Program | Markers |
|---|---|
| proliferation | MKI67, TOP2A, STMN1, TYMS, RRM2, TK1, PCNA, PCLAF, NUSAP1 |
| interferon/cytokine | ISG15, IFIT1, IFIT2, IFIT3, IFI44L, MX1, STAT1, IRF7, IL2RA, IL7R, CCR7, IRF4 |
| T cell | CD3D, CD3E, CD3G, TRAC, TRBC1, TRBC2, TCF7, SELL, LEF1 |
| B cell | MS4A1, CD79A, CD79B, BANK1, BLK, FCRL1, IGHM, IGKC |
| myeloid | LYZ, LST1, S100A8, S100A9, C5AR1, CLEC7A, MS4A7, MAFB |
| erythroid | HBB, HBA1, HBA2, HBD, HBM, AHSP |
| lung/stromal-like | DOCK4, SLC8A1, FMN1, PLXDC2, SLC1A3, NHSL1, LHFPL2, LRP1, NRP1 |

# 5. Amina NK Taxonomy Biomarker Reference

The taxonomy reference is parsed into subtype and state entries. Each entry can have CORE, SUPPORT, CONTEXT, and expected-low/negative markers. The agent uses these hits to rank candidate NK subtype/state labels.

## Allowed structured label layers

| | |
|---|---|
| **Allowed NK subtype calls** | NK2, NK1, adaptive_NK_CMV, adaptive_NK_nonCMV, trNK, cNK, L6_Developmental_immature, Non-NK |
| **Allowed NK state calls** | Chemokine_inflammatory, Checkpoint_exhausted, ER_stress_UPR, Metabolic_stress_hypoxia, Proliferating, IFN_stimulated, Cytotoxic_activated, Homeostatic_quiescent, CIML_cytokine_preactivated, CIMP_cytokine_primed_memory_like, Non-NK |

## Taxonomy scoring

• DE genes used for taxonomy matching are selected from the all-marker Wilcoxon table when available.

• Positive taxonomy evidence: adjusted P < 0.05, logFC > 0.25, pct in cluster >= 0.1.

• Negative/expected-low evidence: adjusted P < 0.05, logFC < -0.25, pct in reference >= 0.1.

• Up to 100 taxonomy DE genes are kept per cluster, ranked by absolute logFC then adjusted P value.

• Taxonomy marker score: CORE=4, SUPPORT=2, CONTEXT=1. Negative-defining markers positive in the cluster are treated as contradictions.

• Support levels: strong, moderate, weak, contradictory, or none, using the thresholds encoded in taxonomy_reference.py.

## Amina taxonomy subtype entries

| Label | Core | Support | Context | Expected low / negative |
|---|---|---|---|---|
| NK2 | GZMK, IL2RB, SELL, XCL1, XCL2, CCR7, NCAM1, IL7R, KLRC1, NCR1 | CXCR3, SPINK2, AREG, GPR183, IKZF2, KIT, KLRK1, LTB, NCR2, TCF7, IL18R1, IL18RAP, PTGDS, FLT3LG | EEF1A1, TNFRSF18, TPT1, S1PR1 | |
| NK1 | CST7, CX3CR1, FCGR3A, FGFBP2, GZMB, NKG7, PRF1, SPON2, GNLY, KLRD1 | ADGRG1, GZMH, GZMM, S1PR5, ZEB2, ACTB, FCER1G, HAVCR2, KLRG1, PRDM1, TBX21, CD160, CHST2, CLIC3, KLF2, LAIR2, SIGLEC7, CFL1, CTSW, GZMA, RAC2 | | |
| adaptive_NK_CMV | KLRC2, B3GAT1, LILRB1, PRDM1, IL32, ZBTB38, FCGR3A | GZMH, CCL5, CX3CR1, TBX21, ZEB2, KLRD1, FGFBP2, SPON2, IFNG, CD244, CD2, FCGR2C, ITGAM | ARID5B, JAKMIP1, PATL2, KIR2DL1, KIR3DL1 | FCER1G, SYK, ZBTB16, KLRC1, KLRB1, CCR7, IL2RB, TYROBP, SIGLEC7 |
| adaptive_NK_nonCMV | PRDM1, IL32, BATF, MAF, CCL5, FCGR3A | GZMH, CD52, ZBTB32, ZBTB38, KIR2DL2, KIR2DL1, KIR3DL1, LILRB1, TMSB4X | CD3E, CX3CR1, TBX21, KLRC2 | |
| trNK | CD69, EOMES, CXCR6, ITGA1 | TBX21, AREG, ITGB1, CCL4L2, HOPX, IKZF3, ITGAE, RGS1, ZNF683 | PDCD1, PSMA2, SCGB1A1, SLC5A3, CLN3 | |
| cNK | CX3CR1 | FCGR3A, S1PR5, KLRG1, KLF2, B3GAT1 | IKZF1 | |
| L6_Developmental_immature | | NCAM1, SPINK2, GATA3, KIT, TCF7, ID2, MYC, CD34, FLT3, LEF1, RUNX2 | IL2RA, IL3RA | |

## Amina taxonomy state entries

| Label | Core | Support | Context | Expected low / negative |
|---|---|---|---|---|
| Chemokine_inflammatory | XCL1, XCL2, CCL4, CCL5, CCL3, IFNG | STAT1, CCL4L2, GZMA, CXCL10, CXCL9 | | |
| Checkpoint_exhausted | TIGIT, HAVCR2, TOX, NR4A2, NR4A1, LAG3, PDCD1, ENTPD1 | KLRC1, CD96, TOX2, BATF, LILRB1, SIGLEC7, KIR2DL1, KIR3DL1, NR4A3, CTLA4, CD274, LAYN, CISH, SOCS1, SOCS3, NFATC1, NFATC2, BHLHE40, IRF4, EZH2, CD38, NT5E, RELB, NFKB2, RXRA, TNFRSF9 | PRDM1, CD244, CD160, KLRG1, CX3CR1, EOMES, IKZF2, ZEB2, MAF, MAFB, EGR2, EGR3, RGS1, BATF3, CDKN1A, HIF1A | |
| ER_stress_UPR | | CD44, CD74, CXCR4, MAFF, ATF3, DDIT3, DNAJB1, EGR2, HSPA1A, HSPA1B, XBP1, EGR3 | DUSP1, GADD45B, BAG3 | |
| Metabolic_stress_hypoxia | | LDHA, MYC, NFE2L2, TXNIP, BNIP3, HIF1A, HK2, MTOR, PDK1, PFKP, SLC2A1 | PRDX1, GCLC, SOD2 | |
| Proliferating | MKI67 | AURKB, CCNB1, CDK1, PCNA, TOP2A, TYMS, UBE2C, BIRC5, CCNB2, STMN1 | | |
| IFN_stimulated | ISG15, MX1, MX2 | BST2, IFI44L, IFIT1, IRF7, OAS1, STAT1, IFIT2, IFIT3 | RSAD2, STAT2, HERC5, OAS3, DDX58, OAS2 | |

| Label | Core | Support | Context | Expected low / negative |
| --- | --- | --- | --- | --- |
| Cytotoxic_activated | FCGR3A, GZMB, NKG7, PRF1, GNLY | CD160, KLRK1, NCR1, CTSW, GZMA, NCR3 | | |
| Homeostatic_quiescent | CST7 | GNLY, KLRD1, CD244, CHST12, IL18RAP, KLRF1 | | |
| CIML_cytokine_preactivated | | GZMB, IL2RB, PRF1, KLRC2, IL12RB2, IFNG, IL12RB1, STAT4 | CD44, CD122, IL15RA | |
| CIMP_cytokine_primed_memory_like | IL2RA, IFNG, IL12RB1, IL12RB2, STAT4, PRF1, GZMB, GZMK, NCR2, NCR3, TCF7 | TNFRSF9, RUNX3, ID2, CD44, NFIL3, JAK2, STAT5A, STAT5B, TNFSF14, LTB, LTA, CSF2, IFNGR2, BCL2, ZBTB32, IRF8, TNF, CD96 | OSM, LIF | CXCR4, S1PR5, NR4A1, NR4A2, NR4A3 |

# 6. How The Annotation Agent Assigns Labels

## Evidence assembled for each cluster

• Top positive and negative cluster-vs-rest DE genes, usually top 50 per direction when available.

• A separate taxonomy-focused DE set selected from the all-marker Wilcoxon table.

• Amina taxonomy marker hits for subtype and state entries.

• Pan-NK marker summary: genes tested, median pan-NK logFC, strongly depleted genes, and median percent nonzero in cluster/reference.

• Non-NK marker summary by lineage group: number of positive markers at logFC >= 0.5 and >= 1.0, strongest non-NK group, and matched genes.

• Lineage sanity check combining non-NK marker signal with whether pan-NK support remains broad or weak.

• Marker program hits for positive and negative DE genes.

• Metadata context: tissue, dataset, and assay composition. This is context, not a label by itself.

• Pairwise DE evidence when available, mainly to split free labels among clusters with the same structured label.

## First-pass annotation prompt rules

• Choose nk_subtype_call from allowed subtype labels or Non-NK.

• Choose nk_state_call from allowed state labels; use NA only when subtype is Non-NK.

• Do not use Unsure; make the best evidence-based call and flag human review when uncertain.

• final_structured_label is training-ready: <subtype>_<state> or Non-NK.

• free_label is a concise biological interpretation and can be more descriptive than the structured label.

• Weak pan-NK DE alone is not evidence for Non-NK because the background is mostly NK/NK-enriched.

• Before assigning an NK subtype, check whether positive marker programs support a non-NK lineage. Strong non-NK markers plus weak/depleted pan-NK support can produce Non-NK.

• T and NK programs can overlap. Shared genes such as GZMK, XCL1, IFNG, CCL3, and CCL4 are not sufficient alone to call T/Non-NK.

## Decision normalization and guardrails

• The LLM output is normalized to allowed subtype/state labels.

• If the model proposes Non-NK but pan-NK markers remain broadly expressed, the graph keeps an NK subtype/state using the best taxonomy match and flags the cluster for review.

• If the subtype/state is missing or invalid, the graph falls back to the best taxonomy-supported subtype/state or a conservative cNK/Homeostatic label and flags review.

• Pairwise split audit can refine free labels when same-structured-label clusters have distinct pairwise DE biology; it does not need to change structured labels.

• The review agent is now anchored on first-pass agent labels/rationale plus marker evidence, without using external/manual annotations.

## Human review flags

• Set when evidence is mixed, lineage is ambiguous, contamination/doublets are plausible, or fallback normalization was needed.

• Examples include strong non-NK markers with residual NK support, T/NK overlap programs, high mitochondrial/stress signatures, epithelial/stromal/myeloid contamination signals, and structured/free-label inconsistencies.

## 7. Practical Interpretation Notes

• Cluster-vs-rest DE is relative to the analyzed background. In this project the background is mostly NK/NK-enriched, so pan-NK depletion often means lower than other NK clusters, not true absence.

• Strong B-cell, myeloid, epithelial/stromal, or erythroid marker dominance can still support Non-NK, especially when pan-NK support is weak.

• T-vs-NK calls require extra caution because activation, chemokine, cytotoxic, and inflammatory genes overlap between T and NK programs.

• The structured label is intended for downstream training; the free label records finer biological interpretation, contamination concern, tissue context, or stress state.

End of report.